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Reserve Estimation Based on Post-Stack Seismic Inversion via Multilayer Perceptron, Random Forest, and Extra Tree Regression Algorithms

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Abstract

This study presents an enhanced reserve estimation approach that mitigates the limitations of conventional inversion methods, thereby facilitating more informed decision-making during exploration. The training process used acoustic impedance (AI) logs as the target variable and depth, two-way travel time, and seismic attributes at the well log scale as inputs. The models employed multilayer perceptrons, random forests, and extra tree regressions to predict the AI, which was then compared to the BLIMP (band-limited impedance inversion) data. The methodology emphasizes selecting suitable cutoffs or thresholds of porosity and permeability for optimal performance. Using cutoff values that match porosity and permeability extremes reduces error and improves reservoir parameter estimation for net pay. The hydrocarbon pore volume is subsequently computed, followed by the estimation of reserves and economic value. Compared with the available cited ground truth data (1.8310e+11 STB and 3.9549e+12 SAR), the ETR results in more accurate reserves and values (1.7174e+11 STB and 3.7096e+12 SAR). These results are also supported by the ETR's empirical value ($a=0.1584$) being closest to the value of the ground truth ($a=0.1585$). We conclude that, compared with the ground truth, the ML models exhibit varying performance, with the accurate trend of $ETR > RFR > MLPR$. In contrast, the values of the BLIMP are smaller than those of the ETR, provided that the threshold choices meet the optimum criteria. The novelty of this study lies in the introduction of a new way of obtaining reserve estimation based on the ML approach of poststack seismic inversion with good accuracy. Seismic attributes are treated as multiple seismic traces, which serve as inputs to ML models to provide predictive information for reserve estimation.

Keywords: Acoustic impedance, Seismic attributes, MLPR, RFR, ETR, Hydrocarbon reserve estimation

Introduction

In the oil and gas industry, accurate reserve estimation is crucial for field development planning, investment decisions, and production forecasting, as it directly influences production strategies and investment decisions (Cheng et al., 2008). The estimation of reserves in carbonate reservoirs presents significant challenges due to limited core coverage, complex pore systems, and geological heterogeneity (Beneke et al.,

1997). Conventionally, estimation relies on petrophysical information from core and well log data, which are often interpolated across the field using geostatistical techniques such as kriging or sequential Gaussian simulation by incorporating uncertainty in petrophysical parameters (Kok & Ulker, 2007).

Seismic inversion, particularly poststack inversion, has become an essential tool for bridging the resolution gap between seismic and well data by providing continuous estimates of geophysical properties (Veecken et al., 2004). It enables the derivation of important geophysical properties such as acoustic impedance, which can be correlated with reservoir properties such as porosity, permeability, and net pay. However, translating inverted seismic acoustic impedance data into accurate reserve estimates remains a nontrivial task because of the nonlinear and multivariate nature of the relationship (Cerny & Bartel, 2007). One of the widely used poststack inversion methods is band-limited impedance inversion (BLIMP), which often has limitations in inverting acoustic impedance data because of its limited frequency coverage. This results in a bias inversion of the reservoir properties for reserve estimation.

New developments in machine learning (ML) have provided powerful alternatives for modeling such complex relationships. Supervised learning algorithms can be trained on seismic and acoustic impedance log data to predict reservoir properties and reserves from the inputs of seismic-derived attributes. Among these ML models, multilayer perceptron regression (MLPR), random forest regression (RFR), and extra tree regression (ETR) have shown potential because of their ability to handle high-dimensional, nonlinear data and provide robust predictions of acoustic impedance.

This study presents a comparative analysis of the MLPR, RFR, and ETR algorithms for reserve estimation in a carbonate reservoir setting compared with poststack seismic inversion, especially with the BLIMP data. The aim is to evaluate the predictive performance of these models, which highlights the strengths and limitations of each approach and offers insights into their practical deployment in reserve and monetary value modeling workflows. By leveraging machine learning with the input features, including seismic attributes, this study aims to exhibit predictive acoustic impedance for reserve and economic estimation accuracy.

Methodology

Figure 1 illustrates the methodology employed to compute the reserve estimation using the ML approach for the three models compared with BLIMP. The details of the extended method are highlighted in yellow.

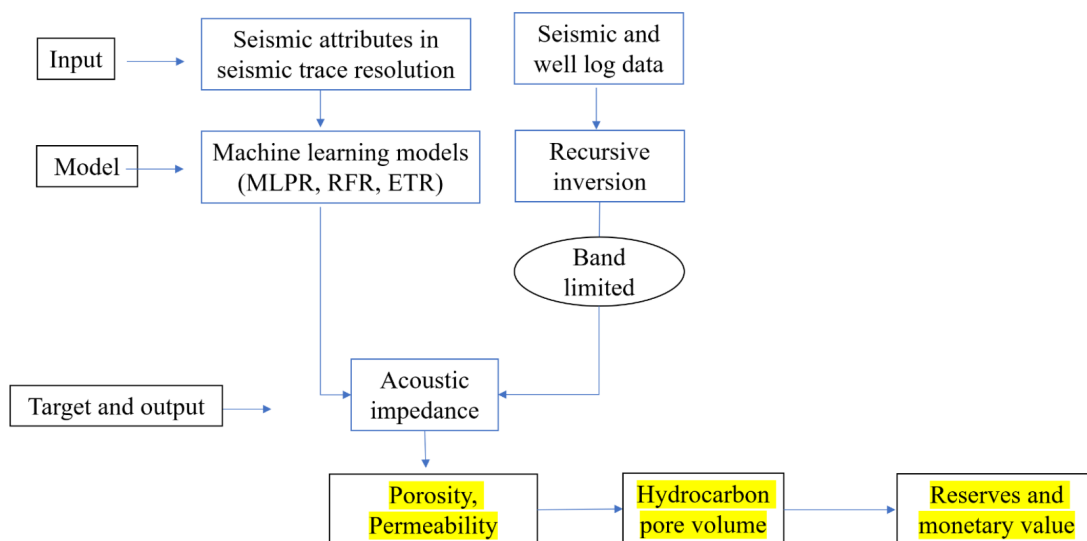


Figure 1—Comparative studies between ML-based and non-ML-based AI inversion for further reserve and monetary estimation.

Dataset, preprocessing, and preexisting work

The dataset for this study was based on preexisting seismic attributes, acoustic impedance logs, and BLIMPs on a well log scale. This dataset was previously prepared with good processing, such that no noise or outliers were present. This dataset was processed with a neural network algorithm (ANN) by [Mardani & Thrust \(2020\)](#) via the TensorFlow library to predict the acoustic impedance based on the inputs of seismic attributes, namely, seismic trace, the integral, second derivative, trace gradient, gradient magnitude, instantaneous frequency, instantaneous phase and cosine of the phase.

Improved machine learning models

The improved results of the acoustic impedance prediction were obtained in our previous publication, in which three supervised models were implemented. The MLPR was chosen to calibrate the ANN method of [Mardani & Thrust \(2020\)](#). Once R^2 was the same, further improvement was obtained via RFR and ETR. The ETR was the most predictable outcome in terms of the determination coefficient (R^2), root mean square error (RMSE), and mean absolute error (MAE). The results of the prediction are shown in the order of ground truth (true AI) > ETR AI > RFR AI > MLPR AI > BLIMP AI ([Surachman et al., 2024](#)).

Reservoir parameter calculation

This study processed the available inverted and predicted acoustic impedance data to compute an optimized form of porosity and permeability via the following relationships.

- Porosity ([Fackler et al., 2012](#); [Avseth et al., 2010](#))

$$\Phi(AI) = a \cdot e^{(-b \cdot AI)} \quad (1)$$

- Permeability ([Herron, 2005](#); [Adam et al., 2018](#))

$$K(\phi) = c \cdot (\phi)^d \quad (2)$$

Cutoff or thresholds for microporous formation of the Illam carbonate reservoir porosity ($\phi_{th}=14.2\%$) and permeability ($K_{th}=0.0205$ mD) obtained from [Khodaei et al., \(2021\)](#) data were selected to find the optimum fitting parameters on the basis of the following objective function.

$$O(a, b, c, d) = \sum_{x \in X} (x_{th} - \min(x))^2 + (x_{th} - \max(x))^2, \text{ Where } X = \{\phi, k\} \quad (3)$$

Net pay determination and volumetric estimation

In this study, net pay or productive pay zones refer to the portion of the reservoir that is porous and permeable enough to produce hydrocarbons economically. For this specific condition, pay zones are flagged if the optimized porosity and optimized permeability are greater than their respective thresholds ([Zeyghami & Taghizadeh Sarvestani, 2023](#)).

Volumetric estimation:

Once we can identify the net pay intervals, we can estimate the hydrocarbon pore volume (HCPV) using the following volumetric formula ([Dake, 1983](#)).

$$HCPV_{net\ pay} = \phi_{net\ pay} (1 - S_w) \cdot A \cdot h \quad (4)$$

where:

- $\phi_{net\ pay}$ = porosity within pay zones (%).
- S_w = average water saturation (%).
- A = drainage area of the reservoir (in m²)
- h = reservoir thickness (final - initial depths) in meters.

This HCPV equation computes the effective pore volume available for hydrocarbon accumulation by removing the water-saturated portion of the pore space from the total porosity.

The oil reserves R in stock tank barrel (STB) are then computed from the HCPV in m^3 by accounting for the formation volume factor B_0 , which defines the ratio of the reservoir volume to the surface volume (Oyeyemi et al., 2019):

$$R = \frac{\text{HCPV}}{B_0 * 0.158987 \frac{m^3}{bbl}} \frac{m^3}{\frac{bbl}{STB}} \quad (5)$$

where:

B_0 is the formation volume factor (in reservoir **bbl/STB**). Note that the reservoir volume is measured in bbl, standing for **barrels** previously known as blue barrels for shipping oil, and the surface standard condition volume is measured in STB, standing for stock tank barrels. The conversion of 1 barrel (bbl) = 42 US gallons = 158.987 liters = 0.158987 m^3 is based on Appendix 2 of Volcanoes: Crucibles of Change (Fisher et al., 1997). STB = 1 bbl measured under standard surface conditions (60°F, 14.7 psi) (Sun & Zhang, 2020).

Finally, **the monetary value in the SAR** of the reserves is determined via:

$$\text{Value} = R * P_o \quad (6)$$

where:

P_o is the unit oil price (in SAR per bbl or SAR/STB).

This calculation provides a simplified yet effective economic monetary evaluation of the reservoir segment based on identified net pay or pay zones and the HCPV. The above workflow enables the comparison of reserve volumes and potential monetary value across the BLIMP, the three different machine learning models, and the ground truth by applying the same geophysical and economic criteria.

Results and discussion

AI Inversion and Petrophysical Property Estimation

The inverted acoustic impedances (AIs) derived from poststack seismic data using the three supervised ML models (MLPR, RFR, and ETR) have different characteristics from the ground truths of the acoustic impedance log and the BLIMP results. Figure 2 shows these variations across depth. To obtain reservoir properties from these well log scale AI values, an optimization-based empirical modeling framework was implemented. The workflow translated AI into porosity and permeability via physically motivated equations (1) and (2). The coefficients a , b , c , and d were optimized for each model's AI prediction via the L-BFGS-B algorithm from `scipy.optimize.minimize` library. The objective of the optimization in equation (3) was to ensure that the predicted and optimized porosity and permeability values spanned ranges closely aligned with those predefined thresholds for net pay classification in the microporous facies of the Illam carbonate reservoir in Section 2.3. This objective function penalizes models whose porosity or permeability predictions deviate significantly from the defined thresholds, which enforces tight bounds around the desired reservoir quality indicators. This is crucial for robust net pay or pay zone determination and reserve calculations.

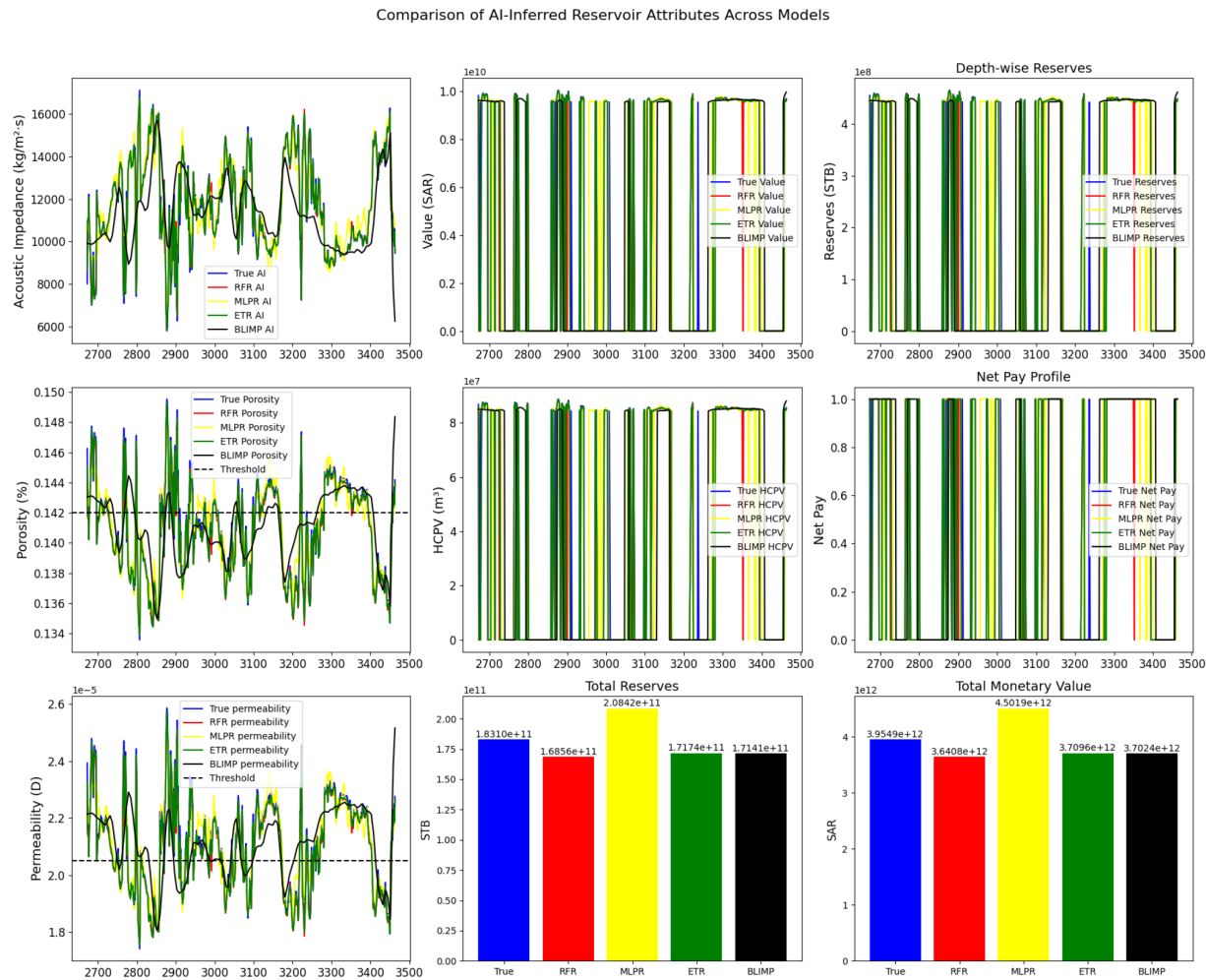


Figure 2—Reserve estimation based on the acoustic impedance-porosity and permeability transformation.

Each AI dataset (from the MLPR, RFR, ETR, BLIMP, and true AI log) was subjected to this optimization procedure independently. The resulting empirical parameters are as follows:

The optimized empirical parameters (Table 1) show strong consistency across all the models, with only minor variations in parameter a. The ETR value of $a = 0.1584$ closely matches the true model ($a = 0.1585$). This confirms that its acoustic impedance predictions align well with realistic porosity behavior. This close calibration also ensures that the ETR downstream outputs, which are net pay identification, HCPV calculation, reserve, and monetary value estimation, prove physically meaningful and economically reliable across models.

Table 1—Optimized parameters for porosity and permeability.

	a	b	c	d
True	0.1585	0.000010	0.0200	3.5000
RFR	0.1582	0.000010	0.0200	3.5000
MLPR	0.1588	0.000010	0.0200	3.5000
ETR	0.1584	0.000010	0.0200	3.5000
BLIMP	0.1580	0.000010	0.0200	3.5000

Net Pay, HCPV, and Reserve Identification

Net pay or pay zone Classification was performed by applying simultaneous thresholds on porosity and permeability, with values previously mentioned in Section 2.3 and conditions in Section 2.4. As shown in the net pay profile (bottom-right plot in Figure 2, the ETR and MLPR models produce longer and more stable pay zone intervals, closely resembling the continuity of the true model. These results support the reliability of their net pay classification. In comparison, the RFR model displays more frequent changes in net pay across depth. For example, at depths of approximately 3300-3400, the RFR has short red bars, indicating sensitivity to localized variations in porosity and permeability. The BLIMP porosity model averages broader productive intervals, which is evident in its consistently conservative net pay output.

The hydrocarbon pore volume (HCPV) Shown in the middle-center panel of Figure 2, is highest for the MLPR model, exceeding $8.5 \times 10^7 \text{ m}^3$ at notified depths of approximately 3300-3400. While MLPR seems not only to predict higher porosity values, it also consistently produces optimized estimations of porosity values that slightly exceed the threshold across the broader range of depths, where more depth intervals exceed the thresholds. This **overestimates** the total net pay throughout the reservoir thickness h in the HCPV calculation in equation (4), resulting in a higher cumulative pore volume. The ETR closely follows the HCPV magnitude, whereas RFR and BLIMP yield lower HCPV values, which is consistent with their characteristic porosity and permeability predictions.

The depthwise reserve distributions In the top-right panel confirm the trend that the ETR and MLPR models show robust reserve continuity across depth, while RFR shows some inconsistent behavior, and BLIMP remains consistently conservative. This is due to the same net pay and HCPV analysis resulting in the same characteristics of reserve distributions as we use the proportional relationship of equation (5). The permeability estimates shown in the bottom-left panel of Figure 2 further support these observations. MLPR and ETR predict porosity and permeability values that more consistently stay above the threshold across multiple depth intervals, contributing to more continuous net pay classification (1, not 0). In contrast, RFR occasionally falls below the porosity threshold, disrupting continuity at the notified depth, whereas BLIMP consistently predicts higher porosity and permeability values, resulting in higher qualifying net pay zones (BLIMP > RFR). These outcomes are a direct result of the parameter optimization performed during AI-to-porosity and permeability conversion. The consistency of the MLPR and ETR across porosity, permeability, and net pay translates into reliable volumetric estimates and lays a solid foundation for subsequent economic evaluation.

Compared with the ground truth (1.8310×10^{11} STB and 3.9549×10^{12} SAR), the ETR model produces the most accurate reserve and monetary value estimates (1.7174×10^{11} STB and 3.7096×10^{12} SAR). This performance is further established by the close match between the ETR's optimized empirical parameters of ($a = 0.1584$, $b = 0.000010$, $c = 0.0200$, $d = 3.5000$) and those derived from the true AI log of ($a = 0.1585$, $b = 0.000010$, $c = 0.0200$, $d = 3.5000$).

While MLPR yields higher absolute reserve and value estimates, RFR predictions are quantitatively closer to the ground truth. The net pay zone highlights the difference between geological continuity (favored by the MLPR) and numerical proximity (favored by the RFR). Compared with those of the MLPR, although the RFR porosity and permeability predictions are noisier in terms of spatial distribution, their total reserve and monetary value estimates deviate less from the ground truth, suggesting improved overall convergence.

Thus, when ranked by proximity to the true values, the model performance trend in reserve estimation is more accurately represented as follows: True > ETR > RFR > MLPR, whereas BLIMP is slightly lower than the ETR values under this optimum condition.

The reference reservoir and economic parameters used for reserve estimation are shown in Table 2. These parameters were applied uniformly to all the models, ensuring a fair and comparative reserve estimation process across the AI outputs from MLPR, RFR, ETR, BLIMP, and the true AI log-based model.

Table 2—Referenced reservoir parameters and economic considerations for reserve and monetary estimation.

Parameter	Value	References	Description
Sw	0.25	A method to estimate $S_w = 0.25$ in the Illam formation (El Sayed & El Sayed, 2019).	Average water saturation
A	1000000	1 km ² drainage area based on the assumption of geological compartmentalization (isolated sections) (Samimi & Karimi, 2014).	Reservoir drainage area in m ²
Bo	1.2	The Bo of the Iranian carbonate range is 1.10 -1.36 bbl/STB (Naseri et al., 2014).	1 STB of oil measured at the surface conditions corresponds to 1.2 bbl at the reservoir conditions
Po	80*0.27	Oscillating oil price prediction between 60 and 90 USD/barrel until 2028 (APANOVYCH et al., 2024)	SAR/barrel or SAR/STB
h	790.96	Our baseline dataset (Mardani & Thrust, 2020)	Reservoir thickness (depth differences in the x-axis of Figure 2)

Economic Valuation

The total estimated reserves and corresponding monetary values for each model were visualized via dual bar plots in Figure 2, consolidating the cumulative predictions across depth. As shown in the top plot, the MLPR model yields the highest estimated reserves of 2.0842×10^{11} STB. However, RFR, while estimating a lower total volume, provides a reserve prediction of 1.6856×10^{11} STB, which is numerically closer to the true value of 1.8310×10^{11} STB. This distinction highlights an important trade-off: RFR favors statistical proximity to the measured ground truth, whereas MLPR favors spatial continuity and reservoir connectivity. Spatial continuity refers to smooth and consistent predictions across depths, whereas reservoir connectivity refers to how well those predictions link economic zones that can flow oil and gas.

In terms of the economic evaluation of Figure 2's bottom plot, the MLPR model achieves the highest monetary value of 4.5019×10^{12} SAR, followed by ETR at 3.7096×10^{12} SAR, which is the closest mirror of its ground truth. RFR, despite its lower total estimate, returns 3.6408×10^{12} SAR, which is remarkably close to the ground-truth value of 3.9549×10^{12} SAR. BLIMP shows values slightly lower than ETR in both reserves and valuations because of its conventional low-frequency impedance and petrophysical predictions in this optimized setting.

This dual evaluation of reserves and monetary value suggests that the ETR offers the most balanced model, which can combine spatial fidelity (geological consistency) and accuracy. MLPR is advantageous when structural continuity is prioritized, whereas RFR is suitable for tasks that demand strict numeric alignment with nonlinear datasets, such as well log data scales.

Model Behavior and Uncertainty

The comparative behavior of the models shows different characteristics. MLPR is excellent for delivering smooth and geologically consistent porosity and permeability distributions, which is valuable for visualizing trends across depths. However, this can come at the cost of slight overestimation. RFR, although less smooth, shows closer numerical proximity to true reserves and values. ETR strikes a strong balance, achieving both spatial coherence and high accuracy in reserves and monetary value predictions.

The limitations of BLIMP are tied to its narrow frequency range, making it consistently conservative and less responsive to reservoir details. However, it serves as a baseline for comparison.

Uncertainty was assessed by evaluating the variation in total reserves across methods. The spread was approximately $-6.7\% \left(\frac{1.6856 - 1.8065}{1.8065} \right)$ to $+15.4\% \left(\frac{2.0842 - 1.8065}{1.8065} \right)$ relative to the mean reserve of 1.8065×10^{11} STB. This is generally acceptable for early-stage reservoir screening. The variability across the ML models reflects their differing assumptions, sensitivities to input features, and interpolation behaviors.

Implications for Field Development

These findings have important implications for reservoir development strategies. The ability of the ML models to predict acoustic impedance and convert it into reliable estimates of porosity and permeability and, ultimately, estimate reserves and monetary values enables a more seismic and well log-driven approach to early-stage decision-making. The trade-off between the interpretability of MLPR and the numerical accuracy of RFR should be considered when designing workflows for static modeling and volumetric classification, where hydrocarbons are likely to accumulate.

Owing to its balance between calibration and trend preservation, the ETR is a preferred model for both technical and economic evaluation. Its close alignment with ground-truth properties and estimated reserves offers confidence in full-field applications. This workflow demonstrates that ML-based inversion can reduce reliance on intensive well control and provides a scalable approach for characterizing 3D reservoirs in future extended work.

Someday, in industry, this methodology may support faster and more accurate screening of reservoir potential, improving decision timelines and capital efficiency in field development planning by deploying only seismic attributes.

Conclusion

1. This research demonstrated the potential of machine learning models (MLPR, RFR, and ETR) for reserve estimation on the basis of poststack seismic inversion using inputs of seismic attributes and well-log scale acoustic impedance as the target. Through physics-guided optimization, AI predictions were successfully converted into porosity and permeability estimates, enabling the determination of net pay, HCPV, reserve, and monetary value. The empirical calibration ensured alignment with geological thresholds relevant to microporous Illam carbonate systems.
2. Among the evaluated models, ETR provided the most accurate reserve and value estimates relative to the ground truth because of its strong balance between geological trend preservation and calibration accuracy. RFR, while less smooth, produced reserve estimates numerically closer to true values than did MLPR, which tended to slightly overestimate. MLPR, however, has advantages in spatial continuity and interpretability. BLIMP, constrained by its spectral limitations, consistently underperformed in all the models except for RFR.
3. The consistent ranking of models by closeness to the true values in this study is as follows: True > ETR > RFR > MLPR, and ETR > BLIMP > RFR.
4. This workflow highlights the ability of ML-based inversion methods to support early-phase reservoir exploration and evaluation. **ETR, in particular, stands out as a powerful candidate for scalable deployment in carbonate reservoir settings, especially when seeking to balance geophysical accuracy with true economic decision-making.**

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References

- Adam, A., Swennen, R., Abdulghani, W., Abdlmutalib, A., Hariri, M., & Abdulraheem, A. (2018). Reservoir heterogeneity and quality of Khuff carbonates in outcrops of central Saudi Arabia. *Marine and Petroleum Geology*, **89**, 721–751.
- Apanovych, Y., Rowland, Z., & Borovková, B. (2024). The Impact of Oil Prices on the GDP of V4 Countries. *Acta Montanistica Slovaca*, **29**(1), 1–12.
- Avseth, P., Mukerji, T., & Mavko, G. (2010). Quantitative seismic interpretation: Applying rock physics tools to reduce interpretation risk. Cambridge university press.
- Beneke, M., Buchalla, G., & Dunietz, I. (1997). Mixing induced (CP) asymmetries in inclusive (B) decays. *Phys. Lett.*, **B393**, 132–142.
- Cerney, B., & Bartel, D. C. (2007). Uncertainties in low-frequency acoustic impedance models. *The Leading Edge*, **26**(1), 74–87.
- Cheng, Y., Lee, W. J., & McVay, D. A. (2008). Quantification of uncertainty in reserve estimation from decline curve analysis of production data for unconventional reservoirs. *Journal of Energy Resources Technology*, **130**(4), 043201.
- Dake, L. P. (1983). Fundamentals of reservoir engineering (Vol. 8). Elsevier.
- El Sayed, N. A., & El Sayed, A. M. A. (2019). Hydraulic-Electrical Flow Units and Water Saturation Calculation: Dam Formation, South Western Qatar, Arabian Gulf. *IOP Conference Series: Earth and Environmental Science*, **362**(1), 12002.
- Fackler, C., Dieckman, E., & Xiang, N. (2012). Porous material parameter estimation: A Bayesian approach. *AIP Conference Proceedings 31st*, **1443**(1), 314–321.
- Fisher, R. V., Heiken, G., & Hulén, J. B. (1997). Volcanoes: Crucibles of Change. Princeton University Press.
- Herron, M. M. (2005). US Patent Application US2005/0148080A1: Carbonate Permeability.
- Khodaei, N., Rezaee, P., Honarmand, J., & Abdollahi-Fard, I. (2021). Controls of depositional facies and diagenetic processes on reservoir quality of the Santonian carbonate sequences (Ilam Formation) in the Abadan Plain, Iran. *Carbonates and Evaporites*, **36**, 1–24.
- Kok, M. V., & Ulker, B. (2007). Reserve estimation using geostatistics. *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects*, **30**(2), 93–100.
- Mardani, R. A., & Thrust, G. V. (2020). Estimation of Acoustic Impedance from Seismic Data in Well-log Resolution Using Machine Learning, Neural Network, and Comparison with Band-limited Seismic Inversion. 1–5. <https://geoconvention.com/wp-content/uploads/abstracts/2020/56036-estimation-of-acoustic-impedance-from-seismic-data.pdf>
- Naseri, A., Khishvand, M., & Sheikhlou, A. A. (2014). A correlations approach for prediction of PVT properties of reservoir oils. *Petroleum Science and Technology*, **32**(17), 2123–2136.
- Oyeyemi, K. D., Olowokere, M. T., & Aizebeokhai, A. P. (2019). Prospect analysis and hydrocarbon reservoir volume estimation in an exploration field, shallow offshore depobelt, Western Niger Delta, Nigeria. *Natural Resources Research*, **28**, 173–185.
- Samimi, A. K., & Karimi, G. (2014). SENSITIVITY & UNCERTAINTY ANALYSIS OF ORIGINAL OIL-IN-PLACE IN CARBONATE RESERVOIR MODELING, A CASE STUDY. *Petroleum & Coal*, **56**(3).
- Sun, S., & Zhang, T. (2020). Reservoir simulations: Machine learning and modeling. Gulf Professional Publishing.
- Surachman, L. M., Abdulraheem, A., Al-Shuhail, A., & Kaka, S. I. (2024). Acoustic impedance prediction based on extended seismic attributes using multilayer perceptron, random forest, and extra tree regressor algorithms. *Journal of Petroleum Exploration and Production Technology*, **14**(7), 1923–1931.
- Veeken, P. C. H., & Da Silva, M. (2004). Seismic inversion methods and some of their constraints. *First Break*, **22**(6), 15–38.
- Zeyghami, M., & Taghizadeh Sarvestani, M. (2023). A comprehensive methodology for reservoir cut-off determination. *Journal of Petroleum Exploration and Production Technology*, **13**(7), 1551–1573.